

BoardNet

Board Examination Communication Network

HSC Board Communication Infrastructure for Bangladesh

Cisco Packet Tracer Project Report

VLSM Tree | Full Addressing Table | Six Router Configurations
DNS | DHCP | Web and Email Services | Redundant Routing

Group Information

- Group ID: 2108
- First Member Student ID: 23201417
- Last four digits: 1417
- Two-octet split: 14 and 17

Base Network: 14.17.0.0/16

Address calculation. The first member's student ID is 23201417. Under the BoardNet rule, its last four digits, 1417, are split into 14 and 17, giving the base network 14.17.0.0/16.

Project Overview

BoardNet is a unified digital network for secure coordination among six Higher Secondary Certificate education boards in Bangladesh. It supports examination coordination, result processing, administrative communication, web access, DNS resolution, DHCP, and email.

Table 1: Board Host Requirements

Board	Code	Estimated Hosts
Dhaka Board	DHK	300
Chattogram Board	CTG	200
Rajshahi Board	RAJ	160
Sylhet Board	SYL	150
Barishal Board	BAR	120
Khulna Board	KHU	120

1. Complete VLSM Allocation Tree

The last four digits of the first member's ID are 1417. Splitting them into two two-digit values gives 14 and 17, so the assigned base network is:

Range: 14.17.0.0 to 14.17.255.255

The LANs are allocated from largest to smallest. Separate infrastructure subnets are then assigned for the shared coordination switch and the four router-to-router segments.

1.1 VLSM Allocation

Table 2: Complete VLSM Address Allocation (11 Networks)

Unit	Hosts	Network	Prefix	Usable Range	Broadcast	Usable Hosts
DHK	300	14.17.0.0	/23	14.17.0.1–14.17.1.254	14.17.1.255	510
CTG	200	14.17.2.0	/24	14.17.2.1–14.17.2.254	14.17.2.255	254
RAJ	160	14.17.3.0	/24	14.17.3.1–14.17.3.254	14.17.3.255	254
SYL	150	14.17.4.0	/24	14.17.4.1–14.17.4.254	14.17.4.255	254
BAR	120	14.17.5.0	/25	14.17.5.1–14.17.5.126	14.17.5.127	126
KHU	120	14.17.5.128	/25	14.17.5.129–14.17.5.254	14.17.5.255	126
DHK/RAJ /SYL	3	14.17.6.0	/29	14.17.6.1–14.17.6.6	14.17.6.7	6
DHK/CTG	2	14.17.6.8	/30	14.17.6.9–14.17.6.10	14.17.6.11	2
DHK/KHU	2	14.17.6.12	/30	14.17.6.13–14.17.6.14	14.17.6.15	2
DHK/BAR	2	14.17.6.16	/30	14.17.6.17–14.17.6.18	14.17.6.19	2
KHU/BAR	2	14.17.6.20	/30	14.17.6.21–14.17.6.22	14.17.6.23	2

1.2 VLSM Tree

The eleven required networks are shown across two connected allocation panels: six board LANs in the first panel and five infrastructure networks in the second. The allocations form a binary subnet hierarchy inside 14.17.0.0/21, which is the portion currently used from the assigned 14.17.0.0/16 block. The rest of the /16 remains available for future expansion.

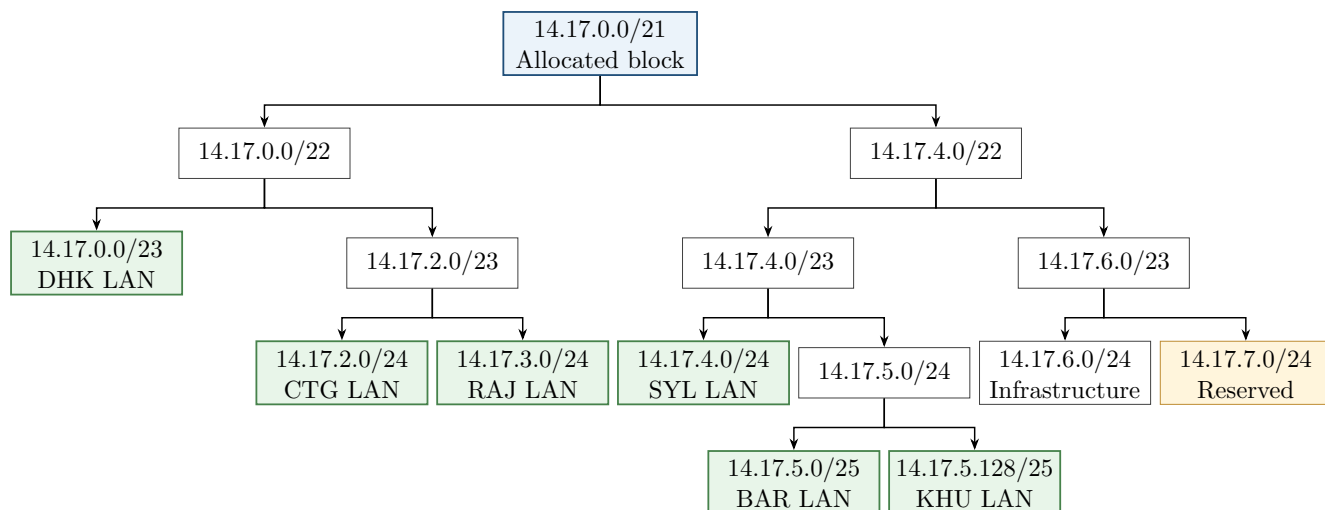


Figure 1: VLSM tree, Part 1: six board LAN networks

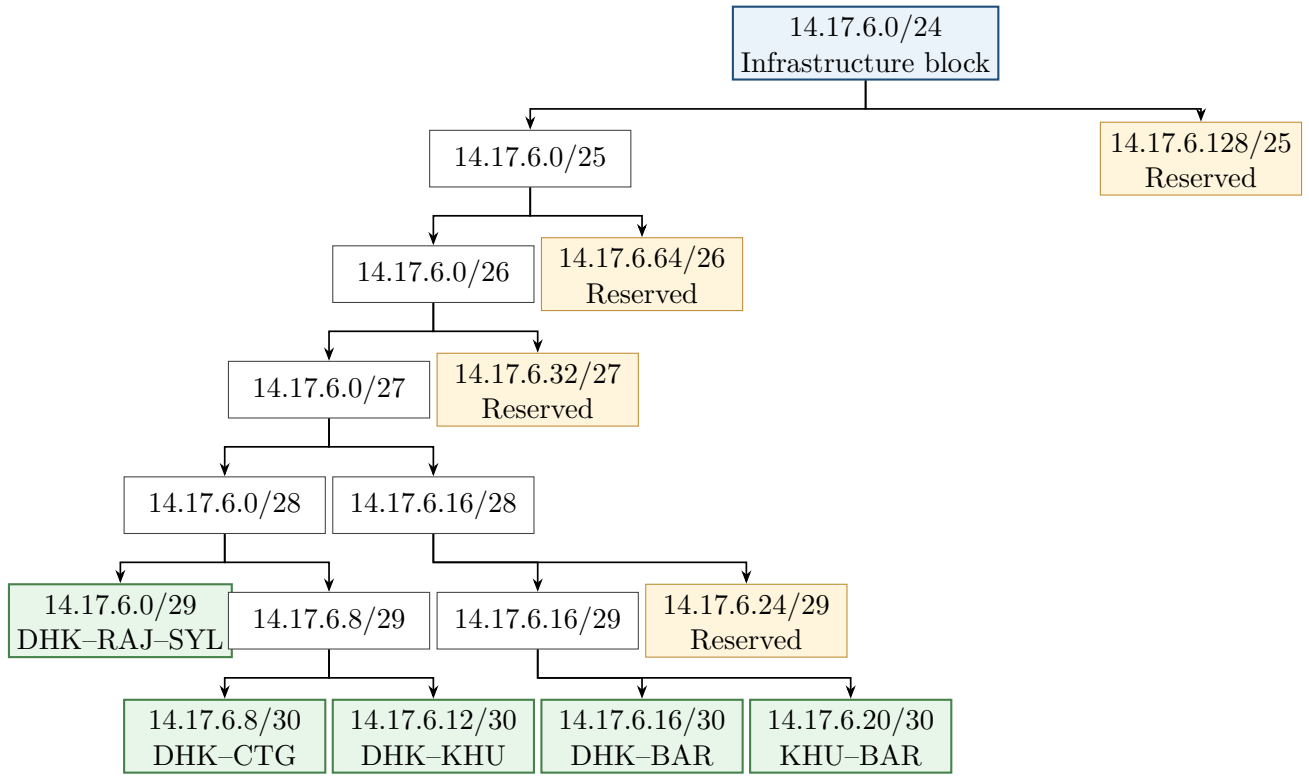


Figure 2: VLSM tree, Part 2: five infrastructure networks

Network Topology

The DHK, RAJ, and SYL routers share a central examination coordination switch. CTG has a single point-to-point connection to DHK. DHK, KHU, and BAR form a triangle so that KHU and BAR retain an alternate route if either direct DHK link fails.

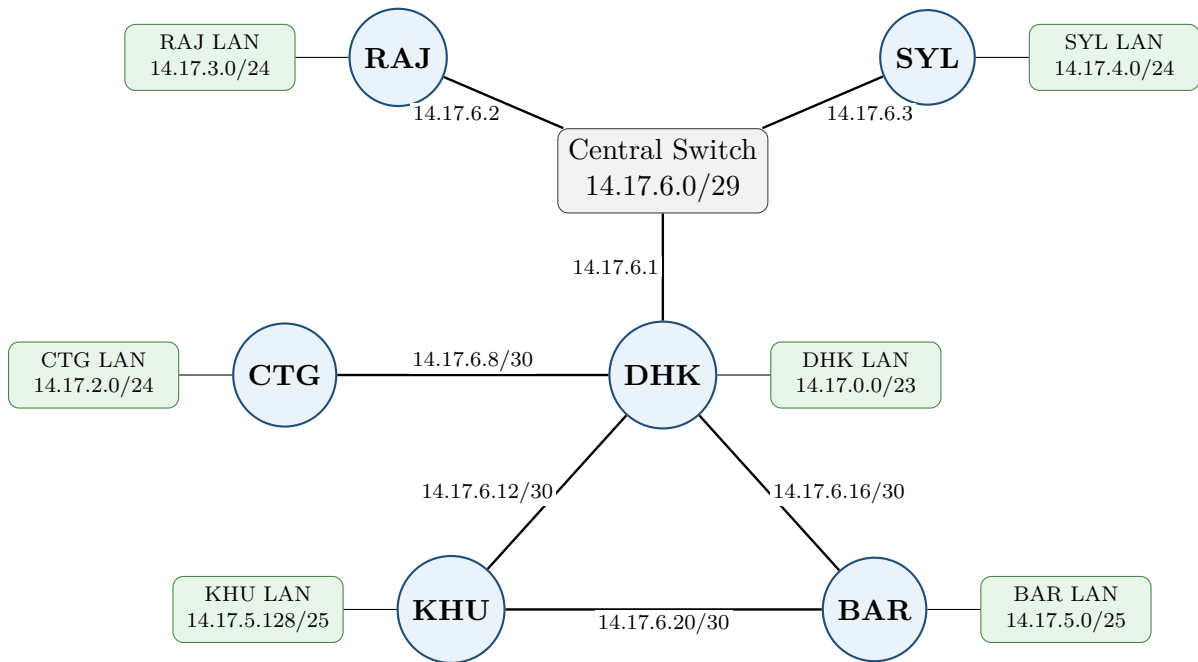


Figure 3: Logical BoardNet topology

2. Full IP Addressing Table

All routers, servers, and printers use manually assigned addresses. Exactly two PCs are shown per board and are configured for DHCP; their final addresses are selected from the listed DHCP ranges.

Table 3: LAN Device Addressing

Board	Device	IP Address	Mask	Gateway
DHK	Router FastEthernet1/0	14.17.0.1	255.255.254.0	–
DHK	Web_DHK	14.17.0.2	255.255.254.0	14.17.0.1
DHK	CENTRAL_DNS _DHK	14.17.0.3	255.255.254.0	14.17.0.1
DHK	Email_DHK	14.17.0.4	255.255.254.0	14.17.0.1
DHK	Printer0	14.17.0.5	255.255.254.0	14.17.0.1
DHK	PC2 and PC3	DHCP: 14.17.0.10–14.17.0.11	255.255.254.0	14.17.0.1
CTG	Router FastEthernet1/0	14.17.2.1	255.255.255.0	–
CTG	Email_CTG	14.17.2.2	255.255.255.0	14.17.2.1
CTG	Printer1	14.17.2.3	255.255.255.0	14.17.2.1
CTG	Web_CTG	14.17.2.4	255.255.255.0	14.17.2.1
CTG	PC4 and PC5	DHCP: 14.17.2.10–14.17.2.11	255.255.255.0	14.17.2.1
RAJ	Router FastEthernet1/0	14.17.3.1	255.255.255.0	–
RAJ	Email_RAJ	14.17.3.2	255.255.255.0	14.17.3.1
RAJ	Printer5	14.17.3.3	255.255.255.0	14.17.3.1
RAJ	PC12 and PC13	DHCP: 14.17.3.10–14.17.3.11	255.255.255.0	14.17.3.1
SYL	Router FastEthernet1/0	14.17.4.1	255.255.255.0	–
SYL	Email_SYL	14.17.4.2	255.255.255.0	14.17.4.1
SYL	Printer4	14.17.4.3	255.255.255.0	14.17.4.1
SYL	PC10 and PC11	DHCP: 14.17.4.10–14.17.4.11	255.255.255.0	14.17.4.1
BAR	Router FastEthernet0/0	14.17.5.1	255.255.255.128	–
BAR	Email_BAR	14.17.5.2	255.255.255.128	14.17.5.1
BAR	DHCP_BAR	14.17.5.3	255.255.255.128	14.17.5.1
BAR	Printer3	14.17.5.4	255.255.255.128	14.17.5.1
BAR	PC8 and PC9	DHCP: 14.17.5.10 onward	255.255.255.128	14.17.5.1
KHU	Router FastEthernet0/0	14.17.5.129	255.255.255.128	–
KHU	Email_KHU	14.17.5.130	255.255.255.128	14.17.5.129
KHU	DHCP_KHU	14.17.5.131	255.255.255.128	14.17.5.129
KHU	Printer2	14.17.5.132	255.255.255.128	14.17.5.129
KHU	PC6 and PC7	DHCP: 14.17.5.138 onward	255.255.255.128	14.17.5.129

Table 4: Router Interconnection Addressing

Segment	Device/Interface	IP Address	Mask
Central switch LAN	DHK FastEthernet0/0	14.17.6.1	255.255.255.248
Central switch LAN	RAJ FastEthernet0/0	14.17.6.2	255.255.255.248
Central switch LAN	SYL FastEthernet0/0	14.17.6.3	255.255.255.248
DHK-CTG	DHK Serial5/0	14.17.6.9	255.255.255.252
DHK-CTG	CTG Serial2/0	14.17.6.10	255.255.255.252
DHK-KHU	DHK Serial2/0	14.17.6.13	255.255.255.252
DHK-KHU	KHU Serial2/0	14.17.6.14	255.255.255.252
DHK-BAR	DHK Serial3/0	14.17.6.17	255.255.255.252
DHK-BAR	BAR Serial3/0	14.17.6.18	255.255.255.252
KHU-BAR	KHU Serial3/0	14.17.6.21	255.255.255.252
KHU-BAR	BAR Serial2/0	14.17.6.22	255.255.255.252

3. Router Configurations

The commands below reproduce the interface names and addressing used in the completed Packet Tracer topology. The serial links used Packet Tracer’s preconfigured clock rate of 2000000, so no manual clock-rate command was required.

3.1 DHK Router

```
enable
configure terminal
hostname DHK
interface FastEthernet0/0
  ip address 14.17.6.1 255.255.255.248
  no shutdown
exit
interface FastEthernet1/0
  ip address 14.17.0.1 255.255.254.0
  no shutdown
exit
interface Serial2/0
  ip address 14.17.6.13 255.255.255.252
  no shutdown
exit
interface Serial3/0
  ip address 14.17.6.17 255.255.255.252
  no shutdown
exit
interface Serial5/0
  ip address 14.17.6.9 255.255.255.252
  no shutdown
exit

router rip
version 2
```

```

no auto-summary
network 14.0.0.0
redistribute static
default-metric 1
exit

ip route 14.17.2.0 255.255.255.0 14.17.6.10
ip route 14.17.5.0 255.255.255.128 14.17.6.18
ip route 14.17.5.128 255.255.255.128 14.17.6.14
ip route 14.17.6.20 255.255.255.252 14.17.6.14
ip route 14.17.6.20 255.255.255.252 14.17.6.18 130

ip dhcp excluded-address 14.17.0.1 14.17.0.9
ip dhcp excluded-address 14.17.2.1 14.17.2.9
ip dhcp excluded-address 14.17.3.1 14.17.3.9
ip dhcp excluded-address 14.17.4.1 14.17.4.9

ip dhcp pool DHK_POOL
network 14.17.0.0 255.255.254.0
default-router 14.17.0.1
dns-server 14.17.0.3
exit
ip dhcp pool CTG_POOL
network 14.17.2.0 255.255.255.0
default-router 14.17.2.1
dns-server 14.17.0.3
exit
ip dhcp pool RAJ_POOL
network 14.17.3.0 255.255.255.0
default-router 14.17.3.1
dns-server 14.17.0.3
exit
ip dhcp pool SYL_POOL
network 14.17.4.0 255.255.255.0
default-router 14.17.4.1
dns-server 14.17.0.3
exit
end
write memory

```

3.2 CTG Router

```

enable
configure terminal
hostname CTG
interface FastEthernet1/0
ip address 14.17.2.1 255.255.255.0
no shutdown
ip helper-address 14.17.0.1
exit
interface Serial2/0
ip address 14.17.6.10 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 14.17.6.9
end
write memory

```

3.3 RAJ Router

```
enable
configure terminal
hostname RAJ
interface FastEthernet0/0
  ip address 14.17.6.2 255.255.255.248
  no shutdown
exit
interface FastEthernet1/0
  ip address 14.17.3.1 255.255.255.0
  no shutdown
  ip helper-address 14.17.0.1
exit
router rip
  version 2
  no auto-summary
  network 14.0.0.0
exit
end
write memory
```

3.4 SYL Router

```
enable
configure terminal
hostname SYL
interface FastEthernet0/0
  ip address 14.17.6.3 255.255.255.248
  no shutdown
exit
interface FastEthernet1/0
  ip address 14.17.4.1 255.255.255.0
  no shutdown
  ip helper-address 14.17.0.1
exit
router rip
  version 2
  no auto-summary
  network 14.0.0.0
exit
end
write memory
```

3.5 KHU Router

```
enable
configure terminal
hostname KHU
interface FastEthernet0/0
  ip address 14.17.5.129 255.255.255.128
  no shutdown
exit
interface Serial2/0
  ip address 14.17.6.14 255.255.255.252
  no shutdown
exit
interface Serial3/0
  ip address 14.17.6.21 255.255.255.252
  no shutdown
exit
```



```

! Primary routes via DHK
ip route 14.17.0.0 255.255.254.0 14.17.6.13
ip route 14.17.2.0 255.255.255.0 14.17.6.13
ip route 14.17.3.0 255.255.255.0 14.17.6.13
ip route 14.17.4.0 255.255.255.0 14.17.6.13
ip route 14.17.6.0 255.255.255.248 14.17.6.13
ip route 14.17.6.8 255.255.255.252 14.17.6.13
ip route 14.17.6.16 255.255.255.252 14.17.6.13
ip route 14.17.5.0 255.255.255.128 14.17.6.22

! Floating backup routes via BAR (distance 130)
ip route 14.17.0.0 255.255.254.0 14.17.6.22 130
ip route 14.17.2.0 255.255.255.0 14.17.6.22 130
ip route 14.17.3.0 255.255.255.0 14.17.6.22 130
ip route 14.17.4.0 255.255.255.0 14.17.6.22 130
end
write memory

```

3.6 BAR Router

```

enable
configure terminal
hostname BAR
interface FastEthernet0/0
 ip address 14.17.5.1 255.255.255.128
 no shutdown
 exit
interface Serial2/0
 ip address 14.17.6.22 255.255.255.252
 no shutdown
 exit
interface Serial3/0
 ip address 14.17.6.18 255.255.255.252
 no shutdown
 exit

! Primary routes via DHK
ip route 14.17.0.0 255.255.254.0 14.17.6.17
ip route 14.17.2.0 255.255.255.0 14.17.6.17
ip route 14.17.3.0 255.255.255.0 14.17.6.17
ip route 14.17.4.0 255.255.255.0 14.17.6.17
ip route 14.17.6.0 255.255.255.248 14.17.6.17
ip route 14.17.6.8 255.255.255.252 14.17.6.17
ip route 14.17.6.12 255.255.255.252 14.17.6.17
ip route 14.17.5.128 255.255.255.128 14.17.6.21

! Floating backup routes via KHU (distance 130)
ip route 14.17.0.0 255.255.254.0 14.17.6.21 130
ip route 14.17.2.0 255.255.255.0 14.17.6.21 130
ip route 14.17.3.0 255.255.255.0 14.17.6.21 130
ip route 14.17.4.0 255.255.255.0 14.17.6.21 130
end
write memory

```

The routes with administrative distance 130 are floating static routes. If the direct DHK–KHU link fails, KHU resolves the alternate next hop through BAR. If the direct DHK–BAR link fails, BAR resolves the alternate next hop through KHU. DHK also has primary and floating routes for the KHU–BAR transit network.

4. DNS Configuration

The central DNS server is `CENTRAL_DNS_DHK` at `14.17.0.3`. Fourteen A records were configured: eight standard hostname records and six bare-domain records used for cross-board SMTP relay.

Table 5: Central DNS Records

Record Name	Type	IP Address	Purpose
<code>www.dhk.edu.bd</code>	A	<code>14.17.0.2</code>	DHK web server
<code>www.ctg.edu.bd</code>	A	<code>14.17.2.4</code>	CTG web server
<code>mail.dhk.edu.bd</code>	A	<code>14.17.0.4</code>	DHK mail hostname
<code>mail.ctg.edu.bd</code>	A	<code>14.17.2.2</code>	CTG mail hostname
<code>mail.raj.edu.bd</code>	A	<code>14.17.3.2</code>	RAJ mail hostname
<code>mail.syl.edu.bd</code>	A	<code>14.17.4.2</code>	SYL mail hostname
<code>mail.khu.edu.bd</code>	A	<code>14.17.5.130</code>	KHU mail hostname
<code>mail.bar.edu.bd</code>	A	<code>14.17.5.2</code>	BAR mail hostname
<code>dhk.edu.bd</code>	A	<code>14.17.0.4</code>	SMTP relay domain
<code>ctg.edu.bd</code>	A	<code>14.17.2.2</code>	SMTP relay domain
<code>raj.edu.bd</code>	A	<code>14.17.3.2</code>	SMTP relay domain
<code>syl.edu.bd</code>	A	<code>14.17.4.2</code>	SMTP relay domain
<code>khu.edu.bd</code>	A	<code>14.17.5.130</code>	SMTP relay domain
<code>bar.edu.bd</code>	A	<code>14.17.5.2</code>	SMTP relay domain

The bare-domain records are required because Packet Tracer resolves the domain portion of a recipient’s email address during server-to-server delivery. DNS resolution was verified both locally and from a remote board.

5. DHCP Configuration

DHK’s router supplies addresses to DHK, CTG, RAJ, and SYL. The `ip helper-address 14.17.0.1` command on the CTG, RAJ, and SYL LAN interfaces forwards client broadcasts to DHK. KHU and BAR use local server devices.

Table 6: Router-Based DHCP Pools

Pool	Network	Gateway	DNS	Excluded Range
DHK_POOL	14.17.0.0/23	14.17.0.1	14.17.0.3	14.17.0.1–14.17.0.9
CTG_POOL	14.17.2.0/24	14.17.2.1	14.17.0.3	14.17.2.1–14.17.2.9
RAJ_POOL	14.17.3.0/24	14.17.3.1	14.17.0.3	14.17.3.1–14.17.3.9
SYL_POOL	14.17.4.0/24	14.17.4.1	14.17.0.3	14.17.4.1–14.17.4.9

Table 7: Dedicated DHCP Server Settings

Server	Own IP	Pool Network	Gateway	DNS	Start Address
DHCP_KHU	14.17.5.131	14.17.5.128/25	14.17.5.129	14.17.0.3	14.17.5.138
DHCP_BAR	14.17.5.3	14.17.5.0/25	14.17.5.1	14.17.0.3	14.17.5.10

On DHCP_KHU and DHCP_BAR, the DHCP service was enabled with the listed pool network, gateway, DNS server, and starting address. The maximum-user values were set within the remaining usable range of each local subnet.

6. Web Server Configuration

Table 8: Web Server Configuration

Server	IP Address	Hostname	Service
Web_DHK	14.17.0.2	www.dhk.edu.bd	HTTP: On
Web_CTG	14.17.2.4	www.ctg.edu.bd	HTTP: On

Both websites were reached by hostname from boards using RIP and from boards using static routing. All clients use 14.17.0.3 as their DNS address.

7. Email Services

Every board has a local email server with SMTP and POP3 enabled. The configured domains, hostnames, and addresses are:

Table 9: Board Email Servers

Server	IP Address	Domain	Hostname	SMTP	POP3
Email_DHK	14.17.0.4	dhk.edu.bd	mail.dhk.edu.bd	On	On
Email_CTG	14.17.2.2	ctg.edu.bd	mail.ctg.edu.bd	On	On
Email_RAJ	14.17.3.2	raj.edu.bd	mail.raj.edu.bd	On	On
Email_SYL	14.17.4.2	syl.edu.bd	mail.syl.edu.bd	On	On
Email_KHU	14.17.5.130	khu.edu.bd	mail.khu.edu.bd	On	On
Email_BAR	14.17.5.2	bar.edu.bd	mail.bar.edu.bd	On	On

For each PC email client, use its full email address, the local mail hostname for both incoming and outgoing servers, and the matching username and password. Cross-board email was verified in both directions between `admin@dhk.edu.bd` and `admin@bar.edu.bd`.

Testing and Verification

Verified Tests

1. DHCP was verified on all twelve PCs. DHK, CTG, RAJ, and SYL received leases from the DHK router; KHU and BAR received leases from their local DHCP servers.
2. Cross-board pings confirmed reachability among routers, PCs, servers, and printers on all six board networks.
3. Local and remote DNS lookups resolved both `www.dhk.edu.bd` and `www.ctg.edu.bd`, including a remote lookup from PC8 on BAR.
4. Both websites opened successfully from RIP boards and statically routed boards.
5. Cross-board email passed in both directions: PC2 (DHK) to PC8 (BAR), and PC8 (BAR) to PC2 (DHK).
6. With DHK Serial2/0 shut down, KHU reached DHK through BAR with 100% success and measured round-trip times of approximately 13–20 ms.
7. With the DHK–BAR link shut down, BAR reached DHK through KHU with 100% success and measured round-trip times of approximately 11–20 ms.

Useful Verification Commands

```
show ip interface brief
show ip route
show ip route static
show ip protocols
show ip dhcp pool
show ip dhcp binding
show running-config | section router rip
show running-config | include ip route
ping 14.17.0.3
ping 14.17.2.4
traceroute 14.17.5.129
```

During redundancy testing, `show ip route` showed the floating route replacing the failed primary route. The successful pings and increased latency confirmed that traffic followed BAR as KHU's

alternate path or KHU as BAR's alternate path.

8. Engineering Assumptions

1. The first member ID is 23201417; therefore, the base network is 14.17.0.0/16.
2. The first usable address of each board LAN is its default gateway.
3. No access control lists were configured because every device must be able to communicate with every other device.
4. Infrastructure addresses below each DHCP starting address are reserved for routers, servers, printers, and future static devices.
5. All ordinary PCs use DHCP; routers, servers, and printers use static addresses.
6. The central switch is a Layer 2 switch and the shared router segment uses 14.17.6.0/29.
7. Point-to-point router connections use /30 networks.
8. DHK, RAJ, and SYL exchange routes using RIPv2 with automatic summarization disabled.
9. DHK redistributes its required static routes into RIP so RAJ and SYL can reach CTG, KHU, and BAR.
10. Bare-domain DNS records such as `bar.edu.bd` are configured in addition to mail hostnames because Packet Tracer resolves the recipient domain during SMTP relay.
11. CTG uses DHK as its only default route and DHCP relay destination.
12. KHU and BAR use local DHCP server devices and explicit static routes.
13. Floating static routes with administrative distance 130 provide the required KHU–BAR recursive backup paths.
14. The topology uses the FastEthernet and Serial interface names shown in the completed Packet Tracer implementation.
15. The DHK–CTG connection uses a Serial DCE/DTE link, consistent with the other point-to-point WAN links in the completed topology.
16. Packet Tracer preconfigured the serial clock rate as 2000000; no manual clock-rate command was required.
17. The central DNS server is 14.17.0.3; only DHK and CTG host web servers.
18. Every board has one email server, one printer, one LAN switch, and exactly two representative PCs shown in the topology.

Summary

BoardNet divides the 14.17.0.0/16 address block efficiently among six education boards. DHK, RAJ, and SYL exchange routes through the central switch using RIPv2; CTG reaches all remote networks through DHK; and KHU and BAR use static routing with reciprocal backup paths. DHK provides centralized DHCP for four boards and central DNS for the entire topology, while KHU and BAR maintain local DHCP servers. Static web, email, DNS, printer, and router addresses make all essential examination services predictable and reachable. Normal operation and both required link-failure scenarios were verified end-to-end in Packet Tracer.